Lecture 5 · Average rates, difference quotients and tangent slopes

20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer; partial completion is welcome.

Q1 · Foundation

Find the slope of the straight line passing through (2,5) and (4,9).

A. 1/2

B. 2

C. 4

D. -2

Source: Jacques 9e, §4.1, Exercise 4.1, 1(a); printed p. 277 / PDF p. 294.

Q2 · Foundation

Find the slope of the straight line passing through (3,-1) and (7,-5).

A. 1

B. -4

C. -1

D. 4

Source: Jacques 9e, §4.1, Exercise 4.1, 1(b); printed p. 277 / PDF p. 294.

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Q3 · Foundation

Find the slope of the straight line passing through (7,19) and (4,19).

A. Undefined

B. -3

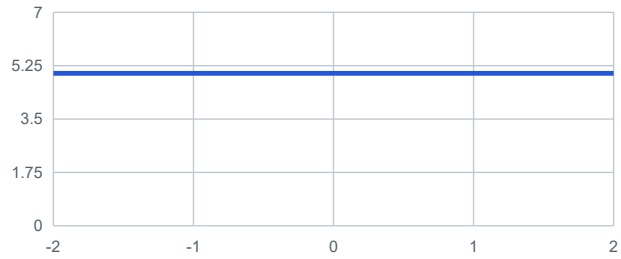
C. 19

D. 0

Source: Jacques 9e, §4.1, Exercise 4.1, 1(c); printed p. 277 / PDF p. 294.

Q4 · Core

For the constant function f(x)=5, which explanation shows why f′(x)=0?



Redrawn textbook graph: f(x) = 5.

A. The graph is horizontal, so its tangent slope is zero.

B. The graph is vertical.

C. The function value is zero.

D. The tangent slope equals the height, 5.

Source: Jacques 9e, §4.1, Exercise 4.1, 3; printed p. 277 / PDF p. 294.

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Q5 · Core

For f(x)=x²-2x, find the function value f(1).

A. 0

B. -1

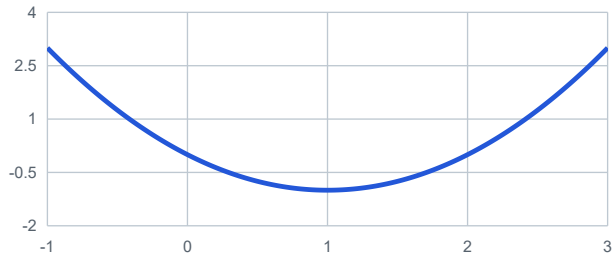
C. 1

D. 2

Source: Jacques 9e, §4.1, Exercise 4.1, 7, table entry x=1; printed p. 277 / PDF p. 294.

Q6 · Core

For f(x)=x²-2x, estimate f′(1) from the tangent at the vertex.



Redrawn textbook graph: f(x) = x² - 2x.

A. -1

B. 1

C. 0

D. 2

Source: Jacques 9e, §4.1, Exercise 4.1, 7(b); printed p. 277 / PDF p. 294.

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Q7 · Application

A teaching position model is $d(t)=t^2+3$ metres, with t in seconds. Choose the correct triple: (position at $t=2$; average speed on $[2,3]$; instantaneous speed at $t=2$).

A. 4 m; 5 m/s; 4 m/s

B. 7 m; 4 m/s; 5 m/s

C. 7 m; 5 m/s; 7 m/s

D. 7 m; 5 m/s; 4 m/s

Source: Jacques 9e, §4.1, Original supplementary delivery model, Lecture 5: $d(t)=t^2+3$; printed p. 268 / PDF p. 285.

Q8 · Application

For $f(x)=x^2-2x$ and $h\neq 0$, simplify $[f(1+h)-f(1)]/h$.

A. h

B. $2h$

C. h^2

D. 0

Source: Jacques 9e, §4.1, Exercise 4.1, 7(b), difference-quotient adaptation; printed p. 277 / PDF p. 294.

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20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer; partial completion is welcome.

Q9 · Application

For the graph $y=\sqrt{(x)}$, A is the point where x=4, and B is the point where x=4.1. Find the gradient of the chord AB.

A. 1/4

B. $(\sqrt{(4.1)}-2)/(4.1)$

C. $(\sqrt{(4.1)}-2)/(0.1)$

D. $\sqrt{(0.1)}$

Source: Jacques 9e, §4.1, Exercise 4.1*, 3(a)(ii); printed p. 278 / PDF p. 295.

Q10 · Challenge (optional)

OPTIONAL. For $d(t)=t^2+3$ metres and $0<h\leq 2$ seconds, find the average speed over [2-h,2+h]. Which statement holds for every permitted h?

A. The average is (4+h) m/s.

B. The average is 4 m/s.

C. The average is (4-h) m/s.

D. The average is 8 m/s.

Source: Jacques 9e, §4.1, Original supplementary symmetric-interval challenge, Lecture 5: optional symmetric measurement; printed p. 268 / PDF p. 285.